Quiz I

Test your knowledge of different systems that you have learned in this course by completing this quiz.

Quiz ?

Statement

A cache is primarily used to improve:

Select Multiple Options

☐ Latency
☐ Concurrency
☐ Fault tolerance
☐ Cost

Check Answer

Quiz ?

Statement

In a distributed environment, a network partition can cause which of the following problems? Select all that apply.

Select Multiple Options

☐ Split-brain
☐ Each of the two servers might think that the other one is dead.
☐ Messages might be sent to the wrong server.
☐ Message or data loss.

Check Answer

Quiz ?

Statement

What is the benefit of using Piggybacked acknowledgments?

Select Multiple Options

☐ It delays the acknowledgment response.
☐ It efficiently uses the available channel bandwidth, thus improving the efficiency of the system.
☐ It decreases the system complexity.
☐ It is an efficient way for the sender to acknowledge the receipt of an acknowledgment.

Check Answer

Quiz ?

Statement

The CAP theorem tells us that:

Select Multiple Options

☐ A distributed system can only have two out of three. Consistency, atomicity, and partition tolerance.
☐ A distributed transaction must have to choose two out the three. Consistency, isolation, and durability.
☐ Any distributed system which is highly-available will have problems with consistency.
☐ In distributed systems, durability and lower latency can only come at the cost of lower consistency.

Check Answer

Quiz ?

Statement

Following is the property of the Write-ahead Log (WAL). Select all that apply.

Select Multiple Options

☐ Data is always sequentially appended to the WAL.
☐ Each log entry in the WAL has a unique identifier.
☐ WAL does not allow updating the data once written.
☐ WAL reduces the read latency by providing a cache containing recent transactions.

Check Answer

Quiz ?

Statement

As opposed to locks, leases:

Select Multiple Options

☐ Have expiration times.
☐ Are long-term while locks are short-term
☐ Can be shared between multiple clients
☐ Are coarse-grained locks

Check Answer

Quiz ?

Statement

What advantage does it give to a system to have separate read and write locks?

Select Multiple Options

☐ Enables concurrent reading of a resource by multiple entities
☐ Enables concurrent writing to a resource by multiple entities
☐ Enables entities to get 'execute' access on a resource if they acquire both read and write locks at the same time
☐ None of the above

Check Answer

Quiz ?

Statement

For GFS, what are the benefits of a large chunk size? Select all that apply.

Select Multiple Options

☐ A small file can all be stored in just one chunk.
☐ Large chunk size provides highly efficient sequential reads and appends of large amounts of data.
☐ A large chunk size simplifies ChunkServer management.
☐ A large chunk size reduces the size of the metadata stored on the master.

Check Answer

Quiz ?

Statement

Consistent hashing ensures that:

Select Multiple Options

☐ Data will be stored on the servers reliably.
☐ If a server is removed, its data will be distributed to all other nodes in the cluster.
☐ Only a small set of keys move when servers are added or removed.
☐ The data will be distributed equally among the servers.

Check Answer

Quiz ?

Statement

Dynamo uses virtual nodes for which of the following reasons? Select all that apply.

Select Multiple Options

☐ To avoid hotspots
☐ To spread the load more evenly across the physical nodes on the cluster by dividing the hash ranges into smaller subranges
☐ To enable the use of virtual machines and virtual memory
☐ To better maintain a cluster containing heterogeneous machines

Check Answer

Quiz ?

Statement

Of the following options, which is not a characteristic of Amazon's Dynamo?

Select Multiple Options

☐ Partition-tolerant
☐ Highly available
☐ Strongly Consistent
☐ Highly reliable

Check Answer

Quiz ?

Statement

Unlike BigTable, with Amazon Dynamo:

Select Multiple Options

☐ Keys are sorted alphabetically to support iteration.
☐ An object is identified by exactly one key.
☐ Two processes may write conflicting updates.
☐ All requests pass through a coordinator.

Check Answer

Quiz ?

Statement

Virtual nodes in Amazon Dynamo are designed to:

Select Multiple Options

☐ Improve fault tolerance due to the replication of nodes
☐ Increase the requests the system can handle since many virtual nodes can be managed by one physical node
☐ Improve load distribution when adding or removing nodes
☐ Create an overlay network that arranges nodes into a logical ring

Check Answer

Quiz ?

Statement

Which of the following are in-memory structures where Cassandra buffers write?

Select Multiple Options

☐ SSTables
☐ NetTables
☐ LinkTables
☐ Memtables

Check Answer

Quiz ?

Statement

What are the disadvantages of using Tombstones in Cassandra? Select all that apply.

Select Multiple Options

☐ Slower reads
☐ Deleted data takes more storage until Tombstones are completely removed during compaction.
☐ Slower writes
☐ Faster writes

Check Answer

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Quiz II

☐ Mark as Completed